Chapter 4. Some Applications of Fourier Series

Isoperimetry, equidistribution, nowhere differentiable functions, heat

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4.1 The isoperimetric inequality

Motivation. In Fourier coefficients, length and area become sums of squares, and Parseval reduces the isoperimetric inequality to $|n| \le n^2$.

Curves. A **parametrized curve** $\gamma : [a,b] \to \mathbb{R}^2$, $\gamma(s) = (x(s), y(s))$, is **simple** if injective on $[a,b)$, **closed** if $\gamma(a) = \gamma(b)$, and C^1 (closed curves $(b-a)$-periodic); the **curve** Γ is its image. Its **length** $\ell = \int_a^b |\gamma'(s)|\, ds$ is invariant under C^1 reparametrization, and if $\gamma' \neq 0$ it can be parametrized by **arc-length**, $|\gamma'| = 1$, $\ell = b - a$. For a simple closed C^1 curve, Green's theorem motivates the definition of the enclosed **area**

$$\mathcal{A} = \frac{1}{2}\Big|\int_\Gamma (x\,dy - y\,dx)\Big| = \frac{1}{2}\Big|\int_a^b (x(s)y'(s) - y(s)x'(s))ds\Big|,$$

Theorem 1.1 (Isoperimetric inequality)

If Γ is a simple closed C^1 curve in $\mathbb{R}^2$ of length ℓ enclosing area $\mathcal{A}$, then $\mathcal{A} \le \frac{\ell^2}{4\pi}$, with equality iff Γ is a circle.

Proof.

Step 1 (normalization). Scaling multiplies ℓ by δ and $\mathcal{A}$ by δ^2, so take $\ell = 2\pi$ and show $\mathcal{A} \le \pi$. With arc-length on $[0, 2\pi]$, $x'^2 + y'^2 = 1$ and

$$\frac{1}{2\pi}\int_0^{2\pi} (|x'(s)|^2 + |y'(s)|^2)ds = 1.$$

Step 2 (Fourier coefficients). With $x \sim \sum a_n e^{ins}$, $y \sim \sum b_n e^{ins}$, we have $x' \sim \sum a_n in e^{ins}$, $y' \sim \sum b_n in e^{ins}$, and Parseval gives

$$\sum_n |n|^2(|a_n|^2 + |b_n|^2) = 1.$$

As x, y are real, $\overline{a_n} = a_{-n}$, $\overline{b_n} = b_{-n}$, and by Lemma 1.5

$$\mathcal{A} = \frac{1}{2}\Big|\int_0^{2\pi} (xy' - yx')\, ds\Big| = \pi\Big|\sum_n n(a_n\overline{b_n} - b_n\overline{a_n})\Big|.$$

Step 3 (the inequality). $|a_n\overline{b_n} - b_n\overline{a_n}| \le 2|a_n||b_n| \le |a_n|^2 + |b_n|^2$ and $|n| \le |n|^2$, so

$$\mathcal{A} \le \pi\sum_n |n|(|a_n|^2 + |b_n|^2) \le \pi\sum_n |n|^2(|a_n|^2 + |b_n|^2) = \pi.$$

Step 4 (equality). $\mathcal{A} = \pi$ forces $a_n = b_n = 0$ for $|n| \ge 2$, so $x = a_{-1}e^{-is} + a_0 + a_1 e^{is}$, $y = b_{-1}e^{-is} + b_0 + b_1 e^{is}$, with $2(|a_1|^2 + |b_1|^2) = 1$. Equality in Step 3 requires $2|a_1||b_1| = |a_1|^2 + |b_1|^2$, so $|a_1| = |b_1| = \frac{1}{2}$, and $|a_1\overline{b_1} - \overline{a_1}b_1| = 2|a_1||b_1|$. With $a_1 = \frac{1}{2}e^{i\alpha}$, $b_1 = \frac{1}{2}e^{i\beta}$: $|\sin(\alpha - \beta)| = 1$, $\alpha - \beta = k\pi/2$, k odd. So $x = a_0 + \cos(\alpha + s)$, $y = b_0 \pm \sin(\alpha + s)$, a unit circle; conversely circles give equality. ∎

4.2 Weyl's equidistribution theorem

Motivation. For irrational γ, averages of $e^{2\pi ikn\gamma}$ are geometric sums tending to 0, so $\langle n\gamma\rangle$ fills $[0,1)$ uniformly.

$[x]$ is the integer part and $\langle x\rangle = x - [x] \in [0,1)$ the **fractional part**; $x = y \bmod \mathbb{Z}$ if $x - y \in \mathbb{Z}$. For rational γ, $\langle n\gamma\rangle$ takes finitely many values; for irrational γ they are distinct ($(n_1 - n_2)\gamma \notin \mathbb{Z}$).

Definition (Equidistribution)

A sequence $\xi_1, \xi_2, \dots$ in $[0,1)$ is **equidistributed** if for every interval $(a,b) \subseteq [0,1)$,

$$\lim_{N\to\infty} \frac{\#\{1 \le n \le N : \xi_n \in (a,b)\}}{N} = b - a.$$

Theorem 2.1 (Weyl)

If γ is irrational, then $\langle\gamma\rangle, \langle 2\gamma\rangle, \langle 3\gamma\rangle, \dots$ is equidistributed in $[0,1)$. In particular $\{\langle n\gamma\rangle\}$ is dense in $[0,1)$ (**Kronecker's theorem**).

With $\chi_{(a,b)}$ the 1-periodic extension of the indicator of $(a,b) \subseteq [0,1)$, $\#\{n \le N : \langle n\gamma\rangle \in (a,b)\} = \sum_{n=1}^{N} \chi_{(a,b)}(n\gamma)$, and the theorem says

$$\frac{1}{N}\sum_{n=1}^{N} \chi_{(a,b)}(n\gamma) \to \int_0^1 \chi_{(a,b)}(x)\,dx. \tag{4.1}$$

Lemma 2.2

If f is continuous, 1-periodic, and γ is irrational, then $\frac{1}{N}\sum_{n=1}^{N} f(n\gamma) \to \int_0^1 f(x)\,dx$.

Proof.

Step 1 (exponentials). $f = 1$ is trivial. For $f = e^{2\pi ikx}$, $k \neq 0$: $\int_0^1 f = 0$ and $e^{2\pi ik\gamma} \neq 1$, so

$$\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ikn\gamma} = \frac{e^{2\pi ik\gamma}}{N}\frac{1 - e^{2\pi ikN\gamma}}{1 - e^{2\pi ik\gamma}}, \qquad \left|\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ikn\gamma}\right| \le \frac{2}{N|1 - e^{2\pi ik\gamma}|} \to 0.$$

By linearity the lemma holds for trigonometric polynomials.

Step 2 (approximation). Corollary 5.4 gives P with $|f - P| < \epsilon/3$; then the averages, the integrals, and (for large N) $\frac{1}{N}\sum P(n\gamma)$ versus $\int P$ each differ by $< \epsilon/3$. ∎

Corollary 2.3

The conclusion of Lemma 2.2 holds for every 1-periodic f that is Riemann integrable on $[0,1]$.

Proof. Let f be real. Take step functions $f_U \ge f \ge f_L$ with $\int(f_U - f_L) < \epsilon$ and modify near the partition points (as in Lemma 3.2) to continuous 1-periodic $f_+ \ge f \ge f_-$ with $\int(f_+ - f_-) < 2\epsilon$. By Lemma 2.2,

$$\limsup \frac{1}{N}\sum f(n\gamma) \le \lim \frac{1}{N}\sum f_+(n\gamma) = \int f_+ \le \int f + 2\epsilon,$$

and similarly $\liminf \ge \int f - 2\epsilon$. ∎

Proof of Theorem 2.1. Apply Corollary 2.3 to $f = \chi_{(a,b)}$, which gives (4.1). ∎

Weyl's criterion (one direction above, the other a book exercise): $\{\xi_n\} \subseteq [0,1)$ is equidistributed iff $\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ik\xi_n} \to 0$ for all $k \neq 0$; e.g. $\langle n^2\gamma\rangle$ for irrational γ. *Geometrically:* a light ray in a square mirror box is periodic for rational slope and equidistributed for irrational slope.

4.3 A continuous but nowhere differentiable function

Motivation. Delayed means isolate one lacunary term, whose derivative is far larger than the $O(\log N)$ that differentiability would allow.

Theorem 3.1 (Weierstrass)

For $0 < \alpha < 1$, $f_\alpha(x) = f(x) = \sum_{n=0}^{\infty} 2^{-n\alpha} e^{i2^n x}$ is continuous but nowhere differentiable.

The series converges absolutely and uniformly ($\sum 2^{-n\alpha} < \infty$), so f is continuous. For nowhere differentiability use the **delayed means**

$$\triangle_N(f) = 2\sigma_{2N}(f) - \sigma_N(f).$$

Facts. (a) As $\sigma_M(f) = \sum_{|n|<M} (1 - \frac{|n|}{M}) a_n e^{inx}$, the coefficient of $a_n e^{inx}$ in $\triangle_N(f)$ is 1 for $|n| \le N$ and $2 - \frac{|n|}{N}$ for $N < |n| < 2N$:

$$\triangle_N(f) = S_N(f) + \sum_{N<|n|<2N} \Big(2 - \frac{|n|}{N}\Big) a_n e^{inx}.$$

(b) For lacunary f (frequencies 2^k) and $N = 2^{n-1}$, no frequency lies strictly between N and $2N$, so $\triangle_N(f) = S_N(f)$.

Lemma 3.2

If g is continuous and differentiable at x_0, then $\sigma_N(g)'(x_0) = O(\log N)$, hence $\triangle_N(g)'(x_0) = O(\log N)$.

Proof.

Step 1. As $\int F_N' = 0$, $\sigma_N(g)'(x_0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} F_N'(t)(g(x_0 - t) - g(x_0))dt$, and differentiability plus boundedness give $|g(x_0 - t) - g(x_0)| \le C|t|$.

Step 2 (bounds for F_N'). $F_N = \sum_{|n|<N} (1 - \frac{|n|}{N}) e^{int}$ gives $|F_N'| \le N^2$; and differentiating $F_N = \frac{\sin^2(Nt/2)}{N \sin^2(t/2)}$ [Supplement],

$$|F_N'(t)| \le \frac{|\sin(Nt/2)\cos(Nt/2)|}{\sin^2(t/2)} + \frac{\sin^2(Nt/2)|\cos(t/2)|}{N|\sin(t/2)|^3} \le \frac{A}{t^2} + \frac{A}{N|t|^3} \le \frac{2A}{t^2} \quad (|t| \ge 1/N).$$

Step 3. So $|\sigma_N(g)'(x_0)| \le C \int_{|t|\le 1/N} N^2 |t|\, dt + C \int_{1/N \le |t| \le \pi} \frac{2A}{|t|}\, dt = O(\log N)$; and $\triangle_N(g)' = 2\sigma_{2N}(g)' - \sigma_N(g)'$. ∎

Lemma 3.3

If $2N = 2^n$, then $\triangle_{2N}(f) - \triangle_N(f) = 2^{-n\alpha} e^{i2^n x}$.

Proof. By Fact (b), $\triangle_{2N}(f) = S_{2N}(f)$, $\triangle_N(f) = S_N(f)$, and the only frequency in $(N, 2N]$ is 2^n. ∎

Proof of Theorem 3.1. If f were differentiable at x_0, Lemma 3.3 gives $|\triangle_{2N}(f)'(x_0) - \triangle_N(f)'(x_0)| = |i2^{n(1-\alpha)} e^{i2^n x_0}| = (2N)^{1-\alpha}$, while Lemma 3.2 gives $O(\log N)$; but $N^{1-\alpha}/\log N \to \infty$. ∎

4.4 The heat equation on the circle

Motivation. On a ring $u = \sum a_n e^{-4\pi^2 n^2 t} e^{2\pi i n x} = f * H_t$: high frequencies die fast, and H_t is a good kernel.

With $\theta = 2\pi x$, $x \in [0,1)$, and rescaled time, $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $u(x,0) = f(x)$. $u = A(x)B(t)$ gives $\frac{B'}{B} = \frac{A''}{A} = \lambda$; periodicity forces $\lambda = -4\pi^2 n^2$, $A = e^{\pm 2\pi i n x}$, $B = e^{-4\pi^2 n^2 t}$. Superposing,

$$u(x,t) = \sum_{n=-\infty}^{\infty} a_n e^{-4\pi^2 n^2 t} e^{2\pi i n x}, \qquad a_n = \hat{f}(n).$$

For integrable f the a_n are bounded, the series converges with all derivatives for $t > 0$ and solves the equation, and $u(\cdot,t) \to f$ in mean square as $t \to 0$ (Parseval). In convolution form $u = f * H_t$ with the **heat kernel**

$$H_t(x) = \sum_{n=-\infty}^{\infty} e^{-4\pi^2 n^2 t} e^{2\pi i n x}.$$

$\{H_t\}$ is a family of good kernels as $t \to 0$ (Corollary 3.4), so $u \to f$ at points of continuity, uniformly for continuous f. H_t has no elementary closed form; its positivity and concentration follow from Poisson summation (Chapter 5).

4.5 Selected exercises

Motivation. The converse of Weyl's criterion and a uniform version, a non-equidistributed sequence, and mean-square initial values for the heat equation.

Exercise 4.5 (Powers of the golden ratio)

Let $\varphi = \frac{1+\sqrt{5}}{2}$ and $\gamma_n = \langle \varphi^n \rangle$. Prove that $\{\gamma_n\}$ is not equidistributed in $[0,1)$.

Solution.

Step 1. $\psi = \frac{1-\sqrt{5}}{2}$ and φ solve $x^2 = x + 1$, so $U_n = \varphi^n + \psi^n$ satisfies $U_{n+1} = U_n + U_{n-1}$, $U_0 = 2$, $U_1 = 1$: all U_n are integers.

Step 2. $0 < |\psi| < 1$ and $\varphi^n = U_n - \psi^n$: for odd n, $\gamma_n = |\psi|^n$; for even n, $\gamma_n = 1 - |\psi|^n$.

Step 3. With $|\psi|^{n_0} < \frac{1}{4}$, $\gamma_n \in [0, \frac{1}{4}) \cup (\frac{3}{4}, 1)$ for $n \geq n_0$, so

$$\frac{\#\{1 \leq n \leq N : \gamma_n \in (\frac{1}{4}, \frac{3}{4})\}}{N} \leq \frac{n_0}{N} \to 0 \neq \frac{1}{2}.$$

■

Exercise 4.7 (Weyl's criterion, converse direction)

Let $\{\xi_n\} \subseteq [0,1)$ be equidistributed. Prove that $\frac{1}{N}\sum_{n=1}^{N} e^{2\pi i k \xi_n} \to 0$ for every integer $k \neq 0$.

Solution. Write $\#_N(E) = \#\{1 \leq n \leq N : \xi_n \in E\}$.

Step 1 (single points are negligible). For $c = 0$: $\#_N(\{0\}) \leq N - \#_N((0,1))$, $\#_N((0,1))/N \to 1$. For $c > 0$: $\#_N(\{c\}) \leq \#_N((c-\delta, c+\delta))$, ratio $\to 2\delta$. So $\#_N(\{c\})/N \to 0$.

Step 2 (continuous f). For real continuous f, $|f| \leq B$, choose $0 = x_0 < \cdots < x_M = 1$ with $|f(x) - f(x_j)| < \epsilon$ on $[x_{j-1}, x_j]$. Each ξ_n lies in some (x_{j-1}, x_j) or equals some x_j, $j < M$, so

$$\Big|\frac{1}{N}\sum_{n=1}^{N} f(\xi_n) - \sum_{j=1}^{M} f(x_j)\frac{\#_N((x_{j-1}, x_j))}{N}\Big| \leq \epsilon + B\sum_{j=0}^{M-1}\frac{\#_N(\{x_j\})}{N}.$$

Step 3. As $N \to \infty$ the sum tends to the Riemann sum $\sum_j f(x_j)(x_j - x_{j-1})$, within ϵ of $\int_0^1 f$, and the error to ϵ (Step 1). So $\frac{1}{N}\sum_{n=1}^{N} f(\xi_n) \to \int_0^1 f$.

Step 4. Apply Step 3 to $\cos(2\pi kx)$, $\sin(2\pi kx)$, $k \neq 0$. ■

Exercise 4.10 (Uniform averages along an equidistributed sequence)

Let $\{\xi_n\}$ be equidistributed in $[0,1)$ and f continuous and 1-periodic with $\int_0^1 f = 0$. Prove that $\frac{1}{N}\sum_{n=1}^{N} f(x+\xi_n) \to 0$ uniformly in x.

Solution.

Step 1 (exponentials). For $k \neq 0$,

$$\Big|\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ik(x+\xi_n)}\Big| = \Big|\frac{1}{N}\sum_{n=1}^{N} e^{2\pi ik\xi_n}\Big| \to 0$$

by Exercise 4.7, independently of x; by linearity the claim holds uniformly for trigonometric polynomials Q with zero constant term.

Step 2 (approximation). Corollary 5.4 (period 1) gives P with $\sup|f-P| < \epsilon$; its constant term has $|c_0| = |\int_0^1 (P-f)| < \epsilon$, so $Q = P - c_0$ has $\sup|f-Q| < 2\epsilon$.

Step 3. For all x,

$$\Big|\frac{1}{N}\sum_{n=1}^{N} f(x+\xi_n)\Big| \le 2\epsilon + \Big|\frac{1}{N}\sum_{n=1}^{N} Q(x+\xi_n)\Big| < 3\epsilon$$

for $N \ge N_0$, N_0 independent of x by Step 1. ■

Exercise 4.11 (Mean-square initial values for the heat equation)

Let f be integrable on $[0,1]$ (1-periodic) and $u(x,t) = (f * H_t)(x)$. Prove that $\int_0^1 |u(x,t) - f(x)|^2 dx \to 0$ as $t \to 0^+$.

Solution.

Step 1. For $t > 0$, $u(\cdot,t) = \sum_n a_n e^{-4\pi^2 n^2 t} e^{2\pi inx}$ converges uniformly, so $u(\cdot,t) - f$ has coefficients $a_n\big(e^{-4\pi^2 n^2 t} - 1\big)$.

Step 2. By Parseval (Theorem 1.3, period 1),

$$\int_0^1 |u(x,t) - f(x)|^2 dx = \sum_n |a_n|^2 \big(1 - e^{-4\pi^2 n^2 t}\big)^2.$$

Step 3. Each factor lies in $[0,1]$ and tends to 0. With $\sum_{|n|>M} |a_n|^2 < \epsilon$ ($\sum |a_n|^2 = \int_0^1 |f|^2$), the sum is at most $\sum_{|n|\le M} |a_n|^2 (1 - e^{-4\pi^2 M^2 t})^2 + \epsilon < 2\epsilon$ for small t. ■